x_1	x_2	x_3	x_4	x_5	x_6
1	2	1	5	6	6
1	1	2	4	5	6
C_1	C_1	C_1	C_2	C_2	C_2

Initial $W = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$ *Step η = 2*

$\Rightarrow C_1$ positive, C_2 negative, Zero is unaccepted

Check classes of $\begin{pmatrix} Z_1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} Z_1 \\ 7 \\ 6 \end{pmatrix} \begin{pmatrix} Z_1 \\ 3 \\ 2 \end{pmatrix}$

Upgraded Vectors (From Top)

$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 6 \end{bmatrix}$

C_1 C_1 C_1 C_2 C_2 C_2

Iteration 1

$d(x_2) = \begin{vmatrix} 1 & 2 & 1 \\ & & \\ & & \end{vmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = 6 \quad \Rightarrow$ Keep same Weight $\quad W = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$

$d(x_3) = \begin{vmatrix} 1 & 1 & 2 \\ & & \\ & & \end{vmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = 6 \quad \Rightarrow$ Keep same Weight $\quad W = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$

$d(x_4) = \begin{vmatrix} 1 & 5 & 4 \\ & & \\ & & \end{vmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = 6 \quad \Rightarrow$ Update Weight $\quad W = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - 2 * \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ -8 \end{bmatrix} \Leftarrow$

$d(x_5) = \begin{vmatrix} 1 & 6 & 5 \\ & & \\ & & \end{vmatrix} \begin{bmatrix} 4 \\ -1 \\ 0 \\ -8 \end{bmatrix} = -96 \quad \Rightarrow$ Keep same Weight $\quad W = \begin{bmatrix} 4 \\ -10 \\ -8 \end{bmatrix}$

$d(x_6) = \begin{vmatrix} 1 & 6 & 6 \\ & & \\ & & \end{vmatrix} \begin{bmatrix} 4 \\ -1 \\ 0 \\ -8 \end{bmatrix} = -104 \quad \Rightarrow$ Keep same Weight $\quad W = \begin{bmatrix} 4 \\ -10 \\ -8 \end{bmatrix}$